

STATISTICS

Regression Examples

Birol Aslanyürek

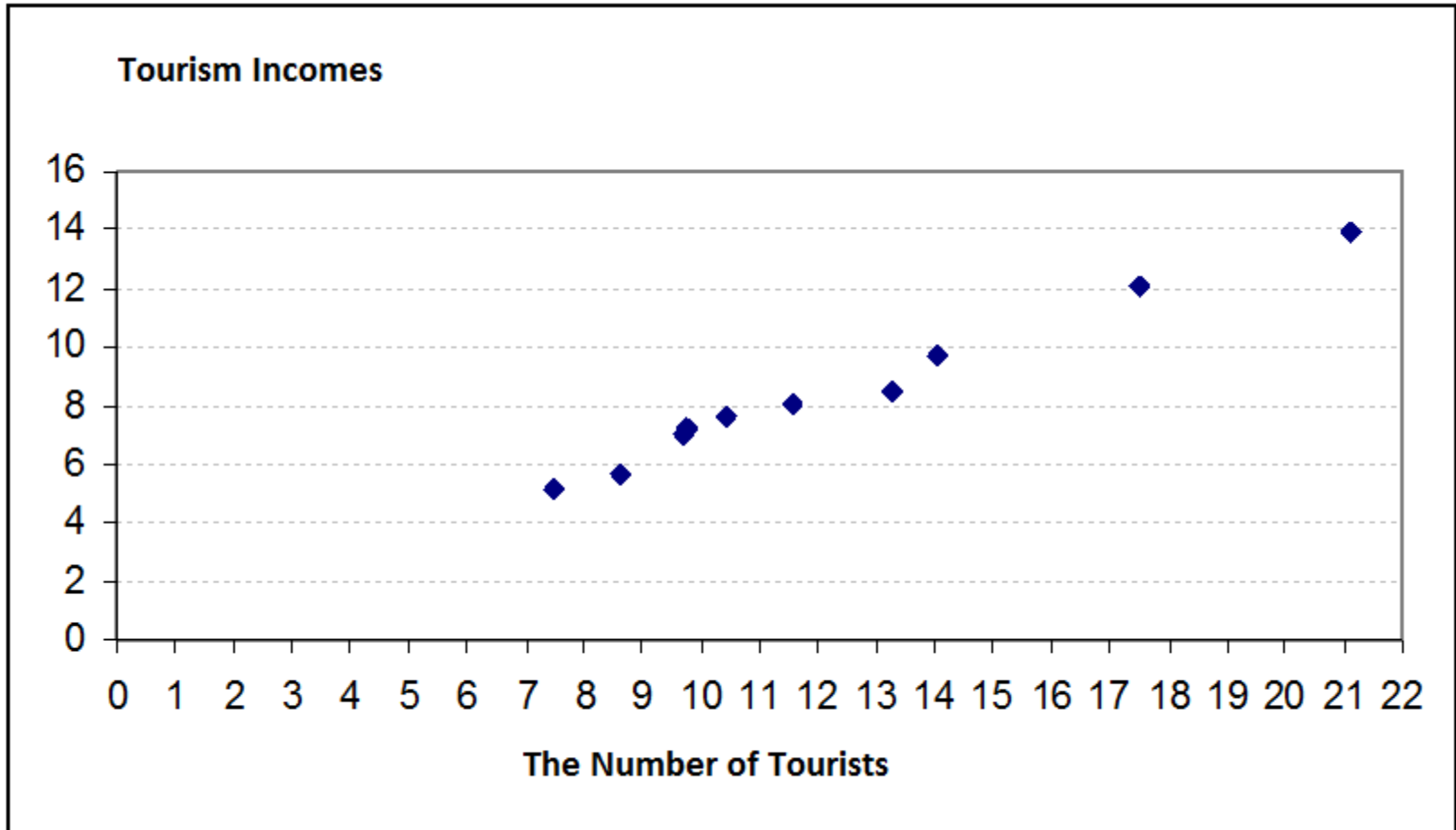
Rev: 03.01.2024

Example: In the years 1996-2005, the tourism incomes of Turkey and the number of tourists who came to Turkey are given below.

Years	Tourism Incomes	The Number of Tourists
1996	5.650	8.614
1997	7.008	9.689
1998	7.177	9.752
1999	5.193	7.464
2000	7.636	10.412
2001	8.090	11.569
2002	8.481	13.247
2003	9.677	14.030
2004	12.125	17.517
2005	13.929	21.122

1. Draw the scatter diagram.
2. Find the estimated regression line.
3. Compute the standard error and variance of the estimator for the estimation of β_0 and β_1 .
4. Compute the variances of the coefficients b_0 and b_1
5. Construct the confidence intervals for β_0 and β_1 ($\alpha=0.05$).
6. Test the hypothesis for $H_0 : \beta_0 = 0$ and $H_a : \beta_0 \neq 0$ ($\alpha=0.05$).
7. Test the hypothesis for $H_0 : \beta_1 = 0$ and $H_a : \beta_1 \neq 0$ ($\alpha=0.05$).
8. Compute SSR, SSE and SST.
9. Compute the determination coefficient and interpret it.
10. Compute the correlation coefficient and interpret it.
11. Test the significance of the regression through the table of variance analysis.
12. Construct the confidence interval of estimation Y_k ($\alpha=0.05$) for $X_k = 8.614$.

1. The Scatter Diagram



2. The Regression Line

Y	X	Y*X	X ²
5.650	8.614	48.6691	74.201
7.008	9.689	67.9005	93.8767
7.177	9.752	69.9901	95.1015
5.193	7.464	38.7605	55.7113
7.636	10.412	79.5060	108.4097
8.090	11.569	93.5932	133.8418
8.481	13.247	112.3478	175.4830
9.677	14.030	135.7683	196.8409
12.125	17.517	212.3936	306.8452
13.929	21.122	294.2083	446.1388
$\Sigma Y=84.966$	$\Sigma X=123.416$	$\Sigma YX=1153.138$	$\Sigma X^2=1686.4501$

2. The Regression Line

$$\begin{aligned}\Sigma Y &= b_0.n + b_1.\Sigma X \\ \Sigma YX &= b_0.\Sigma X + b_1.\Sigma X^2\end{aligned}$$

$$\begin{aligned}84.96 &= b_0.10 + b_1.123.4 \\ 1153.13 &= b_0.123.4 + b_1.1686.4\end{aligned}$$

$$b_0=0.597 \quad b_1=0.640$$

$$\hat{Y} = 0.597 + 0.640X$$

When the number of tourists increases, the amount of tourism income increases.

3. The Standard Error and Variance of the Estimators for the Computation β_0 and β_1

$$s = \sqrt{\frac{\sum e^2}{n-2}}$$

$$s^2 = \frac{\sum e^2}{n-2}$$

Y	Y ²	$\hat{Y} = 0.597 + 0.640X$	$e = Y - \hat{Y}$	e^2
5.65	31.92	$0.597 + 0.640(\mathbf{8.614}) = 6.1099$	-0.460	0.2115
7.008	49.11	$0.597 + 0.640(\mathbf{9.689}) = 6.7979$	0.210	0.0441
7.177	51.51	$0.597 + 0.640(\mathbf{9.752}) = 6.8382$	0.339	0.1147
5.193	26.96	$0.597 + 0.640(\mathbf{7.464}) = 5.3739$	-0.181	0.0327
7.636	58.31	$0.597 + 0.640(\mathbf{10.412}) = 7.2606$	0.375	0.1408
8.09	65.45	$0.597 + 0.640(\mathbf{11.569}) = 8.0011$	0.089	0.0078
8.481	71.93	$0.597 + 0.640(\mathbf{13.247}) = 9.0750$	-0.594	0.3529
9.677	93.65	$0.597 + 0.640(\mathbf{14.030}) = 9.5762$	0.101	0.0101
12.125	147.02	$0.597 + 0.640(\mathbf{17.517}) = 11.8078$	0.317	0.1005
13.929	194.02	$0.597 + 0.640(\mathbf{21.122}) = 14.1150$	-0.186	0.0346
	$\Sigma Y^2 = 789.8721$	$\Sigma \hat{Y} = 84.966$	0.010	$\Sigma e^2 = 1.0501$

$$s = \sqrt{\frac{\sum e^2}{n-k}} = \sqrt{\frac{1.0501}{10-2}} = 0.362$$

$$s^2 = (0.362)^2 = 0.131$$

4. The Variances of the Coefficients b_0 and b_1

$$\text{Var}(b_0) = \frac{\sum X^2}{\sum X^2 - (\sum X)^2/n} S^2 = 0.134$$

$$\text{Var}(b_1) = \frac{1}{n[\sum X^2 - (\sum X)^2/n]} S^2 = 0.00078$$

$$\sqrt{\text{Var}(b_0)} = \sqrt{0.134} = 0.367$$

$$\sqrt{\text{Var}(b_1)} = \sqrt{0.00078} = 0.028$$

$$\hat{Y} = 0.597 + 0.640X$$

5.The confidence intervals for β_0 and β_1 ($\alpha=0.05$)

$$\left[b_0 - t_{\alpha/2, n-2} \sqrt{\text{Var}(b_0)}, \quad b_0 + t_{\alpha/2, n-2} \sqrt{\text{Var}(b_0)} \right]$$

$$[0.597 - 2.306 * (0.367), 0.597 + 2.306 * (0.367)]$$

$$[0.597 - 0.8463, 0.597 + 0.8463]$$

$$0.2493 \leq \beta_0 \leq 1.4433$$

$$\left[b_1 - t_{\alpha/2, n-2} \sqrt{\text{Var}(b_1)}, \quad b_1 + t_{\alpha/2, n-2} \sqrt{\text{Var}(b_1)} \right]$$

$$[0.640 - 2.306 * (0.028) , 0.640 + 2.306 * (0.028)]$$

$$[0.640 - 0.0645, 0.640 + 0.0645]$$

$$0.5755 \leq \beta_1 \leq 0.7045$$

6. Hypothesis Testing for β_0 ($\alpha=0.05$)

$$H_0 : \beta_0 = 0 \quad t_{0.05/2, 8} = 2.306$$

$$H_a : \beta_0 \neq 0$$

$$t_0 = \frac{b_0 - \beta_0}{\sqrt{\text{Var}(b_0)}} = \frac{0.597 - 0}{0.367} = 1.626$$

$$t_0 = 1.626 < t_{0.05/2, 8} = 2.306$$

Accept H_0 .

β_0 is not statistically significant.

7. Hypothesis Testing for β_1 ($\alpha=0.05$)

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

$$t_{0.05/2, 8} = 2.306$$

$$t_1 = \frac{b_1 - \beta_1}{\sqrt{\text{Var}(b_1)}} = \frac{0.640 - 0}{0.028} = 22.85$$

$$t_0 = 22.85 > t_{0.05/2, 8} = 2.306$$

Reject H_0 .

β_1 is statistically significant. There is a linear relationship between X and Y .

8. SST, SSR and SSE

$$\hat{Y} = 0.597 + 0.640X$$

Sum of Total Squares (**SST**) $\sum(Y - \bar{Y})^2 = \sum Y^2 - (\sum Y)^2 / n$

Sum of Regression Squares (**SSR**) $\sum(\hat{Y} - \bar{Y})^2 = b_1^2 \left[\sum X^2 - (\sum X)^2 / n \right]$

Sum of Error Squares (**SSE**) $\sum(Y - \hat{Y})^2 = SST - SSR$

$$\sum X^2 = 1686.4501 \quad \sum X = 123.416 \quad \sum Y = 84.966 \quad \sum Y^2 = 789.8721$$

SST $\rightarrow \sum(Y - \bar{Y})^2 = \sum Y^2 - (\sum Y)^2 / n = 789.8721 - 7219.229 / 10 = 67.9499$

SSR $\rightarrow b_1^2 \left[\sum X^2 - \frac{(\sum X)^2}{n} \right] = (0.640)^2 \left[1686.4501 - \frac{(123.416)^2}{10} \right] = 66.8873$

SSE $\rightarrow SSE = \sum e^2 = 1.0501 \cong 67.9499 - 66.8873$

9. Coefficient Determination R^2 :

$$R^2 = \frac{SSR}{SST} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2} = \frac{66.8873}{67.9499} \cong 0.9844$$

Comment: This result shows that 98.44% of the variation of tourism incomes dependent on the number of tourists. (98.44% of the variability in the data is explained by the regression model)

10. Correlation Coefficient

$$r = \sqrt{0.9844} = 0.9922 \rightarrow$$

$$\hat{Y} = 0.597 + 0.640X$$

There is a positive (since the slope of the regression line is positive) strong relationship between tourism incomes and the number of tourists.

11. Test the significance of the regression through the table of variance analysis.

$$H_0 : \beta_0= \beta_1=0$$

$$H_1 : \beta_0= \beta_1\neq 0$$

<i>Degree of freedom</i>		<i>SS</i>	<i>Average</i>
SSR	1	66.8873	66.8873
SSE	8	1.0501	0.1312
SST	9	67.9499	

$$SSR = (0.640)^2 \left[1686.4501 - \frac{(123.416)^2}{10} \right] = 66.8873$$

$$SSE = \sum e^2 = 1.0501$$

$$SST = \sum (Y - \bar{Y})^2 = \sum Y^2 - (\sum Y)^2 / n = 789.8721 - 7219.229 / 10 = 67.9499$$

$$F_0 = \frac{66.8873}{0.1312} = 509.81 \qquad F_{0.05,1,8} = 5.32$$

$F_0 > F_{0.05,1,8}$; reject H_0 ; at least one of the parameters (β_0 and β_1) is statistically significant.

12. The confidence interval of estimation Y_k ($\alpha=0.05$) for $X_k = 8.614$.

$$\left[\hat{Y}_k - t_{\alpha/2, n-2} S_{\hat{Y}_k}, \quad \hat{Y}_k + t_{\alpha/2, n-2} S_{\hat{Y}_k} \right] \quad S_{\hat{Y}_k} = \sqrt{1 + \frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}} S$$

$$X_k = 8.614$$

$$\hat{Y}_k = 6.1099$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025 \rightarrow t_{0.025, 8} = 2.306$$

$$n - 2 = 8$$

$$6.1099 \pm 2.306 \sqrt{1 + \frac{1}{10} + \frac{(8.614 - 12.3416)^2}{163.2991}} 0.362$$

$$5.20124 \leq Y_k \leq 7.0180$$